Skewness - Measures and Interpretation

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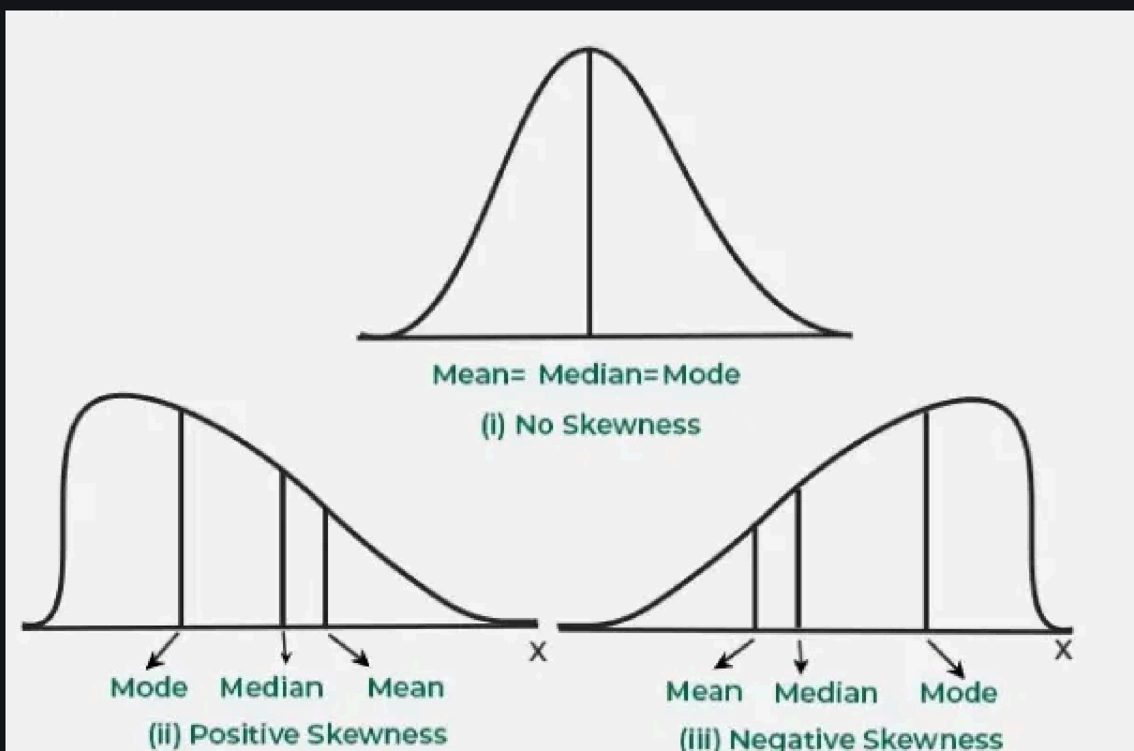
Skewness is a key statistical measure that shows how data is spread out in a dataset. It tells us if the data points are skewed to the left (negative skew) or to the right (positive skew) in relation to the mean. It is important because it helps us to understand the shape of the data distribution which is important for accurate data analysis and helps in identifying outliers and finding the best statistical methods to use for analysis. In this article, we will see skewness, different types of skewness and its core concepts.



Skewness

Types of Skewness

Skewness describes the direction and degree of asymmetry in a dataset's distribution. Various types are as follows:

1. Positive Skewness (Right Skew)

In a positively skewed distribution, the right tail is longer than the left which means most data points are on the left with a few large values pulling the distribution to the right.

Relationship:

$Mean > Median > Mode$

Examples: Income distribution, exam scores and stock market returns.

2. Negative Skewness (Left Skew)

In a negatively skewed distribution, the left tail is longer which means most data points are on the right with a few smaller values pulling the distribution to the left.

Relationship:

$Mean < Median < Mode$

Examples: Test scores on easy exams, age at retirement and gestational age at birth.

3. Zero Skewness (Symmetrical Distribution)

Zero skewness shows a perfectly symmetrical distribution where the mean, median and mode are equal. In a symmetrical distribution, the data points are evenly distributed around the central point.

Relationship:

$Mean = Median = Mode$

Example: A perfectly balanced dataset with equal frequencies of all values.

Tests of Skewness

There are several ways to find the skewness of a dataset which can help to find whether the data is positively skewed, negatively skewed or roughly symmetric. Below are some common methods used to measure skewness: